

# Anshuman Phadke

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## EDUCATION

### Vellore Institute of Technology

Bachelor of Technology Electronics and Communication Engineering

July 2019 – July 2023

CGPA: 9.57/10.0 (Rank Holder)

## TECHNICAL SKILLS

**Programming & Systems:** Python, C, C++, MATLAB, PostgreSQL, Linux, ROS, Docker, Kubernetes, Redis, WebSockets

**AI Frameworks:** PyTorch, TensorFlow, scikit-learn, OpenCV, NumPy, Pandas, CUDA, MLflow, Kubeflow, TensorRT, ONNX

**Models & Methods:** Regression, Classification (SVM, KNN), Clustering (K-Means, DBSCAN), Dimensionality Reduction (PCA), CNNs, RNNs (LSTM/GRU), Transformers, Autoencoders, Diffusion Models, Optimization

**Visualization & Tools:** Plotly, Dash, Folium (OSM), Matplotlib, Seaborn, Flask, FastAPI, QGIS, Git, Jira, Confluence, ChatGPT Codex, GitHub Copilot, Claude Code, AWS Sagemaker, Vector CANoe, Vector CANape, DTS Monaco

## EXPERIENCE

### Research Engineer

Sep 2025 – Present

Drone Research Center - Indian Institute of Technology, Bombay

Mumbai, India

- Developed a cascaded real-time detection and classification pipeline using YOLO-OBb and ByteTrack for multi-object tracking, achieving 0.95 mAP, optimized and deployed on NVIDIA Jetson Orin NX for Indian Army.
- Developing the system architecture for a scalable UAV Traffic Management (UTM) platform for large-scale drone coordination, incorporating 3D airspace corridor modeling, conflict-based multi-agent path planning, optimization, collision avoidance, and flight scheduling under the Maharashtra Drone Mission.
- Built a UTM visualization and analytics platform using Flask, CesiumJS, WebSockets, and Redis, enabling real-time telemetry ingestion for 100 concurrent drones with geospatial querying via PostGIS for dynamic corridor loading and registration workflows.
- Implemented a hardware-in-the-loop (HIL) validation environment integrating NVIDIA Jetson Orin NX with CubePilot Orange flight controllers running PX4, enabling simulation-based evaluation of path-planning algorithms over the IIT Bombay airspace map.

### ADAS Engineer

Aug 2023 – Jul 2025

Mercedes-Benz Research & Development India (MBRDI)

Bengaluru, India

- Developed production-grade radar signal processing software and a Radar Analytics Tool (RAT) enabling multiple ADAS functions, including Blind Spot Monitoring, Active Brake Assist, and Adaptive Cruise Control.
- Developed a CNN-based audio perception system using ResNet-50 and MFCC feature extraction for emergency vehicle detection, classification, and direction-of-arrival estimation.
- Developed internal Python applications using PyQt that reduced manual workflows by approximately 90% and improved engineering productivity, saving nearly \$125,000.
- Filed **8 patents** with the Indian Patent Office in the autonomous driving domain.

### Student Trainee Intern

Sep 2022 – May 2023

Mercedes-Benz Research & Development India (MBRDI)

Bengaluru, India

- Built a scalable simulation framework for autonomous parking validation in Mercedes-Benz vehicles, accelerating virtual verification of ultrasonic sensing performance and improving development efficiency for parking assistance functions.
- Led electromagnetic and ray-tracing based performance studies of automotive ultrasonic sensors using ANSYS HFSS and SBR+, creating automated Python and MATLAB analytics pipelines to benchmark sensor behavior against safety and system-level KPIs across multiple vehicle architectures.

## PROJECTS

### SSTV-Based IoT Data Acquisition and Analytics | [Research Paper](#)

Feb 2023

- Designed an IoT-based data acquisition and analytics system for remote regions, enabling reliable sensing, transmission, and analysis under constrained connectivity.

### Team Assailing Falcons (SAE Aero Design East) | [Website](#)

Dec 2020 – May 2023

- Vice Captain and Propulsion Head of a 14-member team competing in SAE Aero Design East (Advanced Class), achieving **Global Rank 3**, Rank 1 in Technical Presentation, and Rank 1 in Design Report
- Developed a Python-based ground station data acquisition system for real-time avionics telemetry
- Developed a vision based drop zone detection and autonomous navigation system

### ADAMS – Advanced Driver Assistance and Monitoring System | [Project Link](#)

Nov 2020 – Jan 2021

- Built a computer vision and deep learning-based driver monitoring and assistance system with features like: facial recognition, emotion classification, drowsiness detection, head movement.

### AgriHero – Smart Agriculture Platform | [Project Link](#)

April 2021 - June 2021

- Built a distributed IoT sensing and ML platform delivering real-time environmental and agronomic insights to enable data-driven decision support and improved accessibility for rural stakeholders.

### Formula1-2025-Season-Analytics-Dashboard | [Project Link](#)

Dec 2025 - Jan 2026

- A sleek, data driven analytics dashboard for the Formula 1 2025 season delivering deep insights into driver performance, team dominance, race trends and season long comparisons.